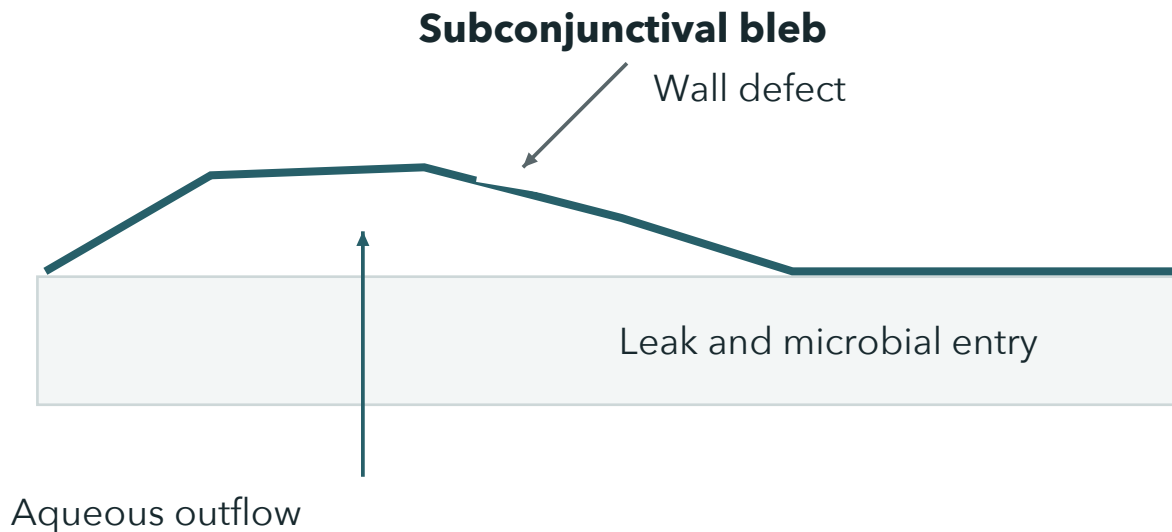


Bleb and tube threats: leakage, infection and exposure

Part 2 | A filtering route can become an entry route

The bleb wall is part of long-term infection safety.



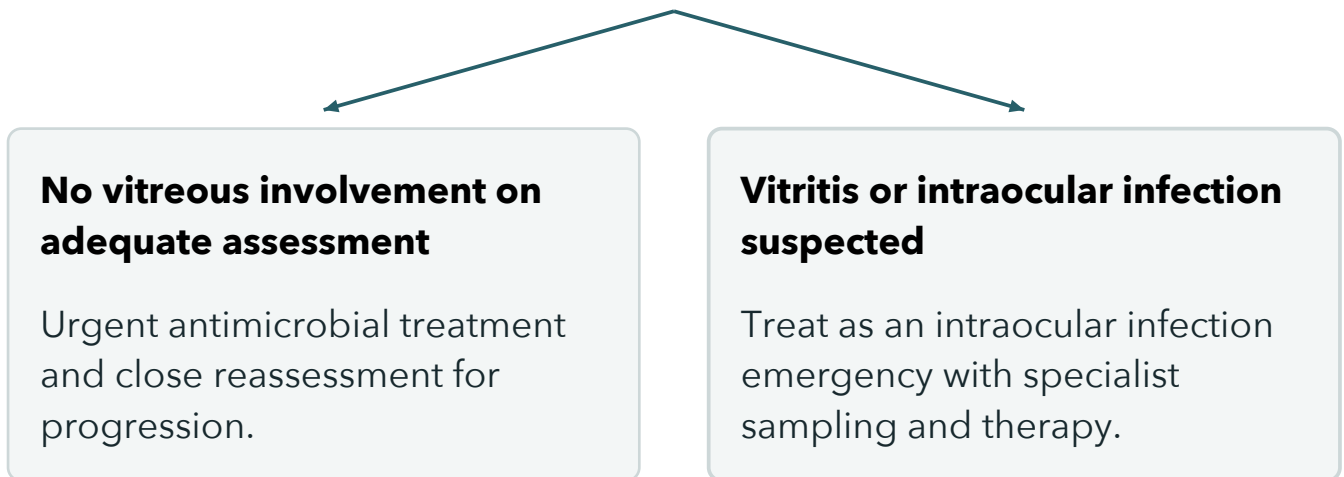
Original section: aqueous exits beneath conjunctiva; a wall defect permits leakage and microbial access.

Inspect bleb vascularity, integrity and surrounding inflammation. A thin leaking bleb deserves assessment even without pain or major pressure loss. Fluorescein testing helps identify leakage, but inflammatory material can obscure it. Long-term success includes tissue safety and surveillance, not just a low IOP.

Blebitis is distinguished by the compartments involved

Vitreous involvement changes the treatment pathway.

Is the infection confined to the bleb?



An opacified inflamed bleb can accompany anterior chamber reaction. Do not wait for severe pain or profound visual loss before assessing the posterior segment. An apparently localized infection can progress; the initial label must be revised if vitreous signs or worsening vision develop.

Tube exposure needs source correction

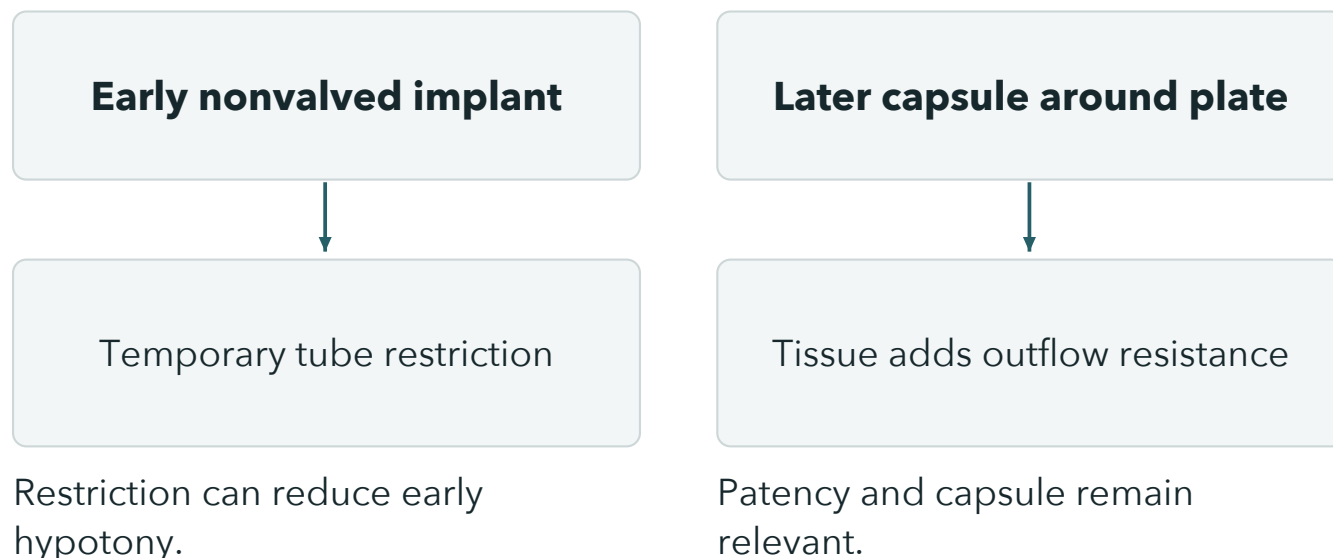
A quiet eye is not evidence that exposed hardware is safe.

Finding	Management implication
Conjunctival erosion over tube	Prompt surgical assessment and repair planning
Discharge or intraocular inflammation	Evaluate for associated infection urgently
Recurrent exposure after repair	Review tissue, tube position and mechanical factors
Pressure controlled	Does not remove infection risk

Exposed drainage hardware creates a microbial route. Antibiotics do not rebuild the tissue barrier. Repair or device modification depends on the defect and ocular status. Worsening redness, pain or vision warrants urgent contact.

Valve status changes the early physiology

A plate capsule later provides resistance that is initially absent.



Original mechanism map: temporary restriction and later tissue resistance are different elements.

Nonvalved devices commonly need temporary restriction to reduce early hypotony. Valved devices supply intrinsic resistance, but pressure complications remain possible. Later IOP depends on capsule resistance, patency and ocular factors. Early hypotony and later encapsulation require different interpretation.

Follow-up should preserve both pressure and tissue

An apparently successful operation can acquire a new risk over time.

Review target	Reason
Surface and bleb or tube integrity	Detect leak or erosion
IOP trend and optic nerve function	Judge efficacy rather than one reading
Corneal health and tube contact	Identify endothelial threat
Patient symptoms and access	Support timely response to infection

Separate loss of pressure control from a safety complication. A device can remain effective while its covering tissue fails, and a bleb can look quiet while leaking. Document changes over time and make the urgent-contact plan concrete. Revision choices should address the identified failure mechanism.